US 20250278912A1

(19) **United States**

(12) **Patent Application Publication**

Yu

(10) **Pub. No.: US 2025/0278912 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **PET RECONSTRUCTION SYSTEM BASED ON A VIRTUAL WORLD INFRASTRUCTURE AND METHOD FOR IMPLEMENTING THEREOF**

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(21) Appl. No.: **19/068,390**

(22) Filed: **Mar. 3, 2025**

(30) **Foreign Application Priority Data**

Mar. 1, 2024 (TW) 113107626

Publication Classification

(51) **Int. Cl.**

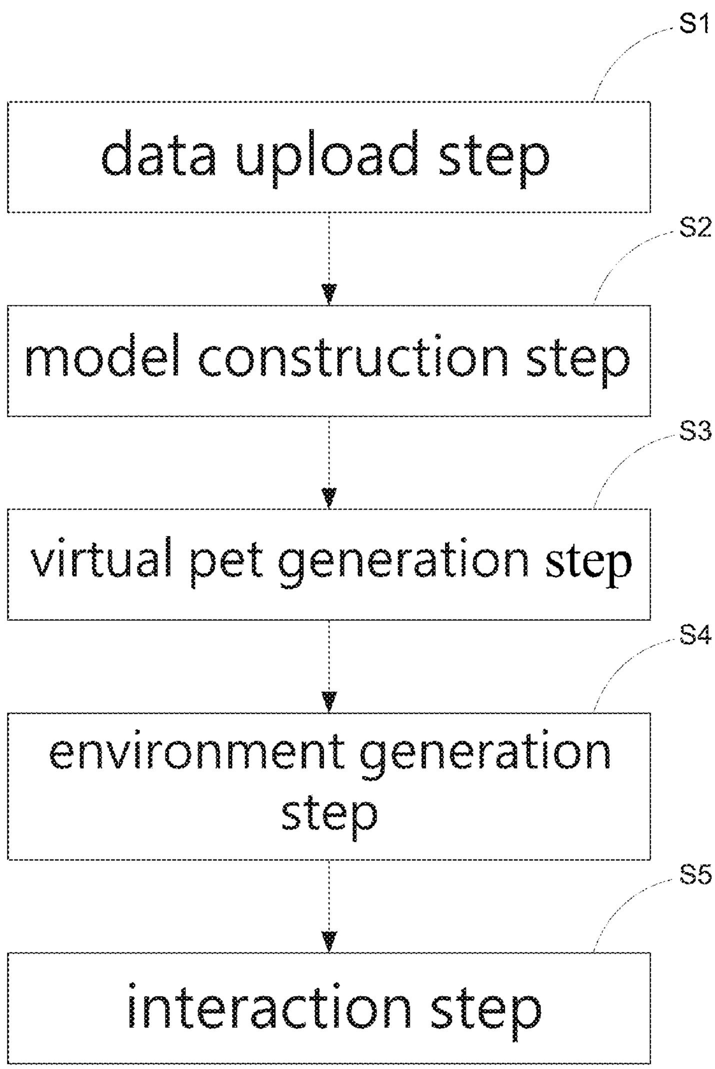
G06T 19/20	(2011.01)
G06T 13/40	(2011.01)

(52) **U.S. Cl.**

CPC ***G06T 19/20*** (2013.01); ***G06T 13/40*** (2013.01); *G06T 2219/2008* (2013.01)

(57) **ABSTRACT**

The present disclosure relates to a pet reconstruction system based on a virtual world infrastructure and its implementation method. The system primarily comprises a server and one or more terminal devices. The server is in data communication with a model database, a customization database, a data processing module, a virtual pet generation module, an environment generation module, and a real-time analysis module. By analyzing the features of a particular animal and comparing them with models stored in the model database, a virtual identity corresponding to the particular animal is generated. Finally, the real-time analysis module receives at least one command, enabling the virtual identity to execute its model based on the received command. This allows real-time adjustment of the virtual pet's behavior, thereby recreating the emotional connection between the pet owner and the pet in a virtual environment.



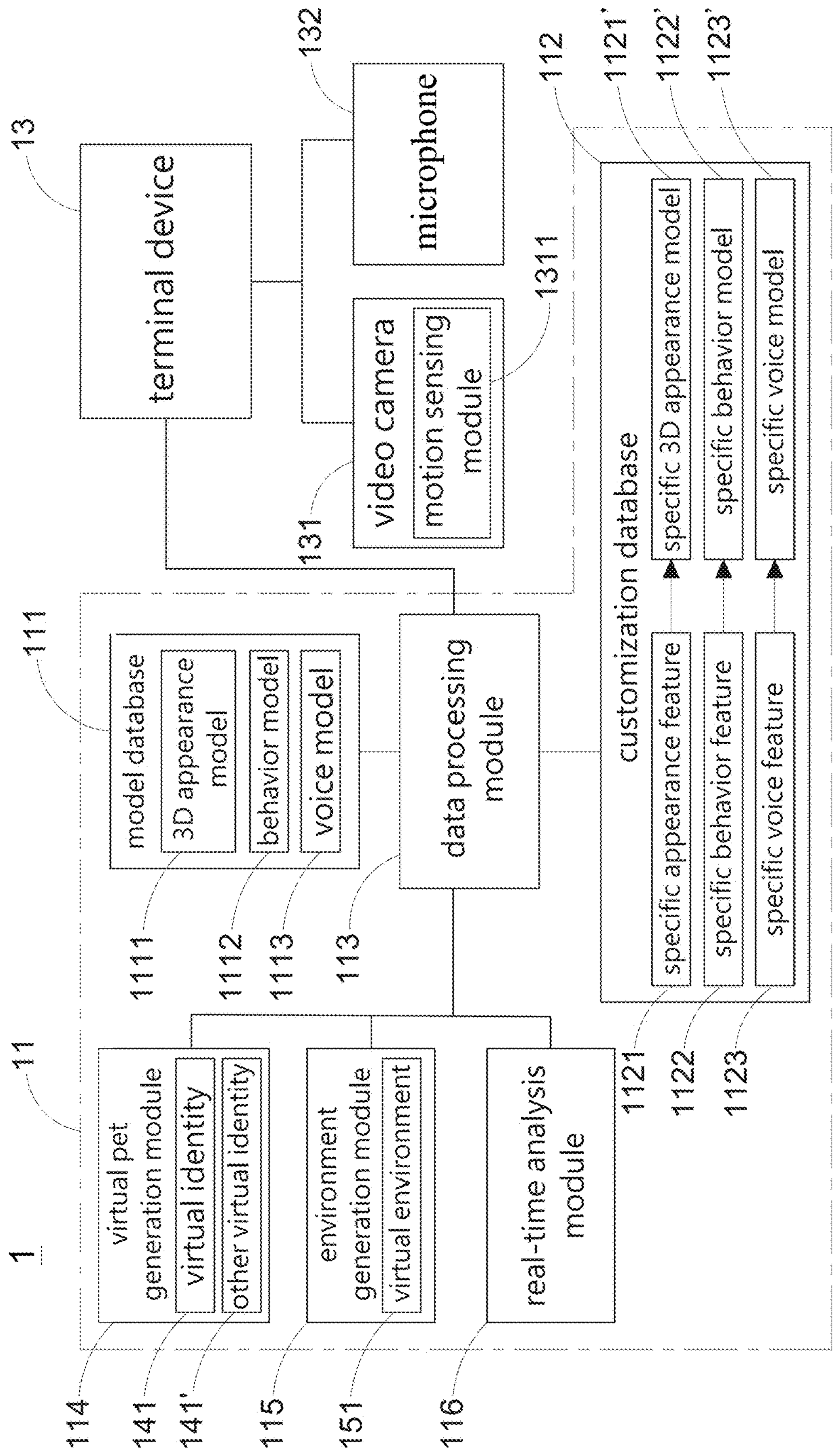

1
11
virtual pet generation module
virtual identity
other virtual identity
environment generation module
virtual environment
real-time analysis module
114
1141
1141'
115
1151
116
111
model database
3D appearance model
behavior model
voice model
1111
1112
1113
data processing module
113
customization database
specific appearance feature
specific behavior feature
specific voice feature
1121
1122
1123
specific 3D appearance model
specific behavior model
specific voice model
112
1121'
1122'
1123'
13
terminal device
video camera
motion sensing module
131
1311
microphone
132

Fig.1

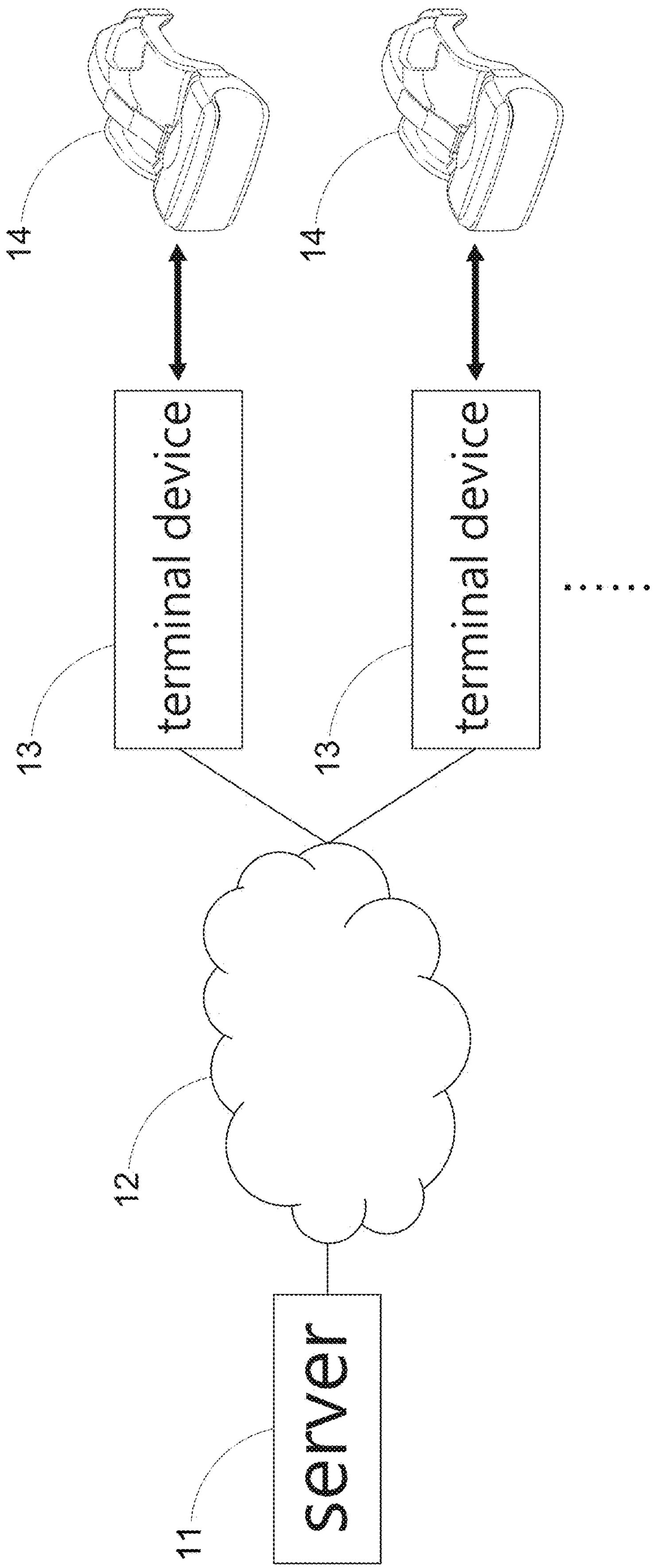


Fig.2

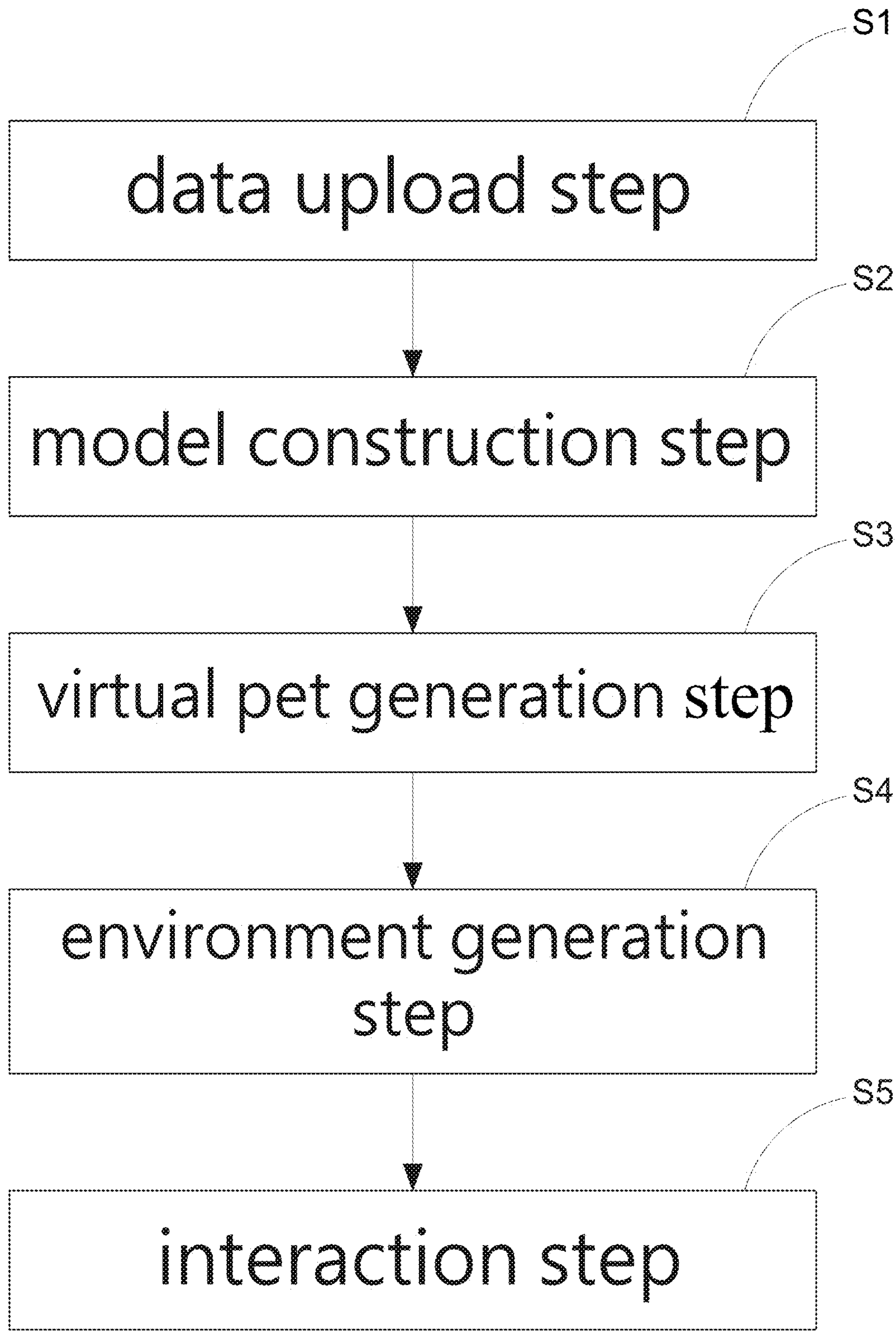

S1
data upload step
S2
model construction step
S3
virtual pet generation step
S4
environment generation step
S5
interaction step

Fig.3

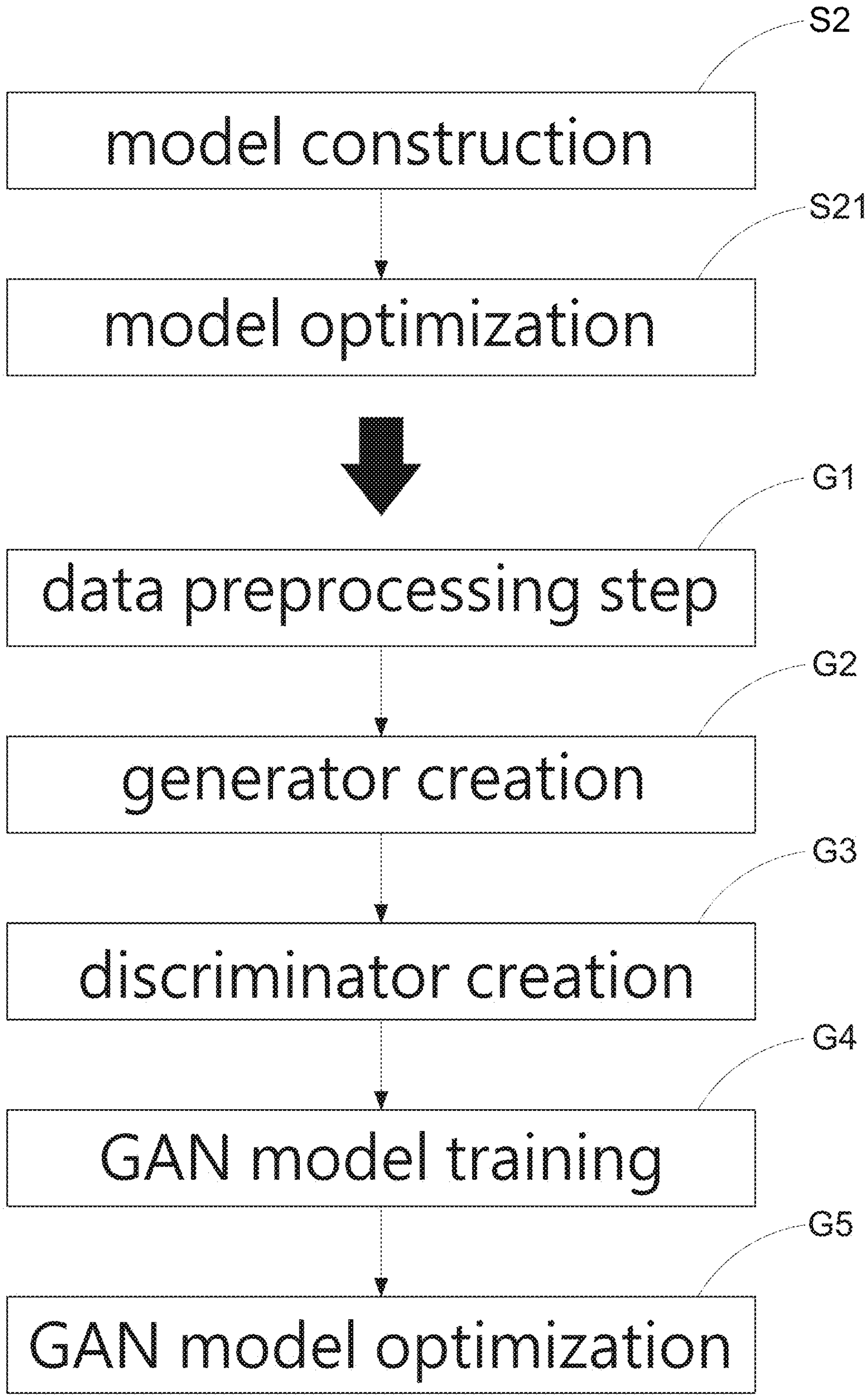

S2
model construction
S21
model optimization
G1
data preprocessing step
G2
generator creation
G3
discriminator creation
G4
GAN model training
G5
GAN model optimization

Fig.4

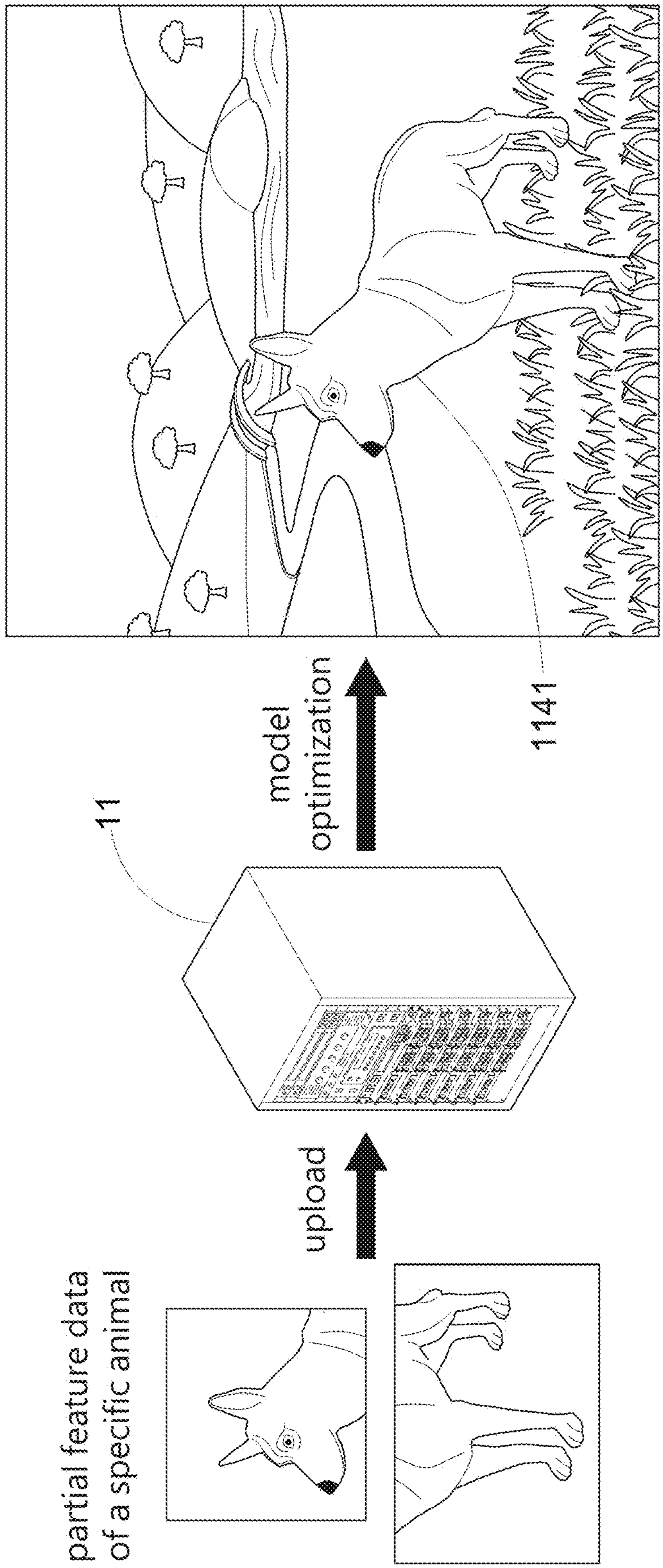

partial feature data
of a specific animal
upload
11
model
optimization
1141

Fig.5

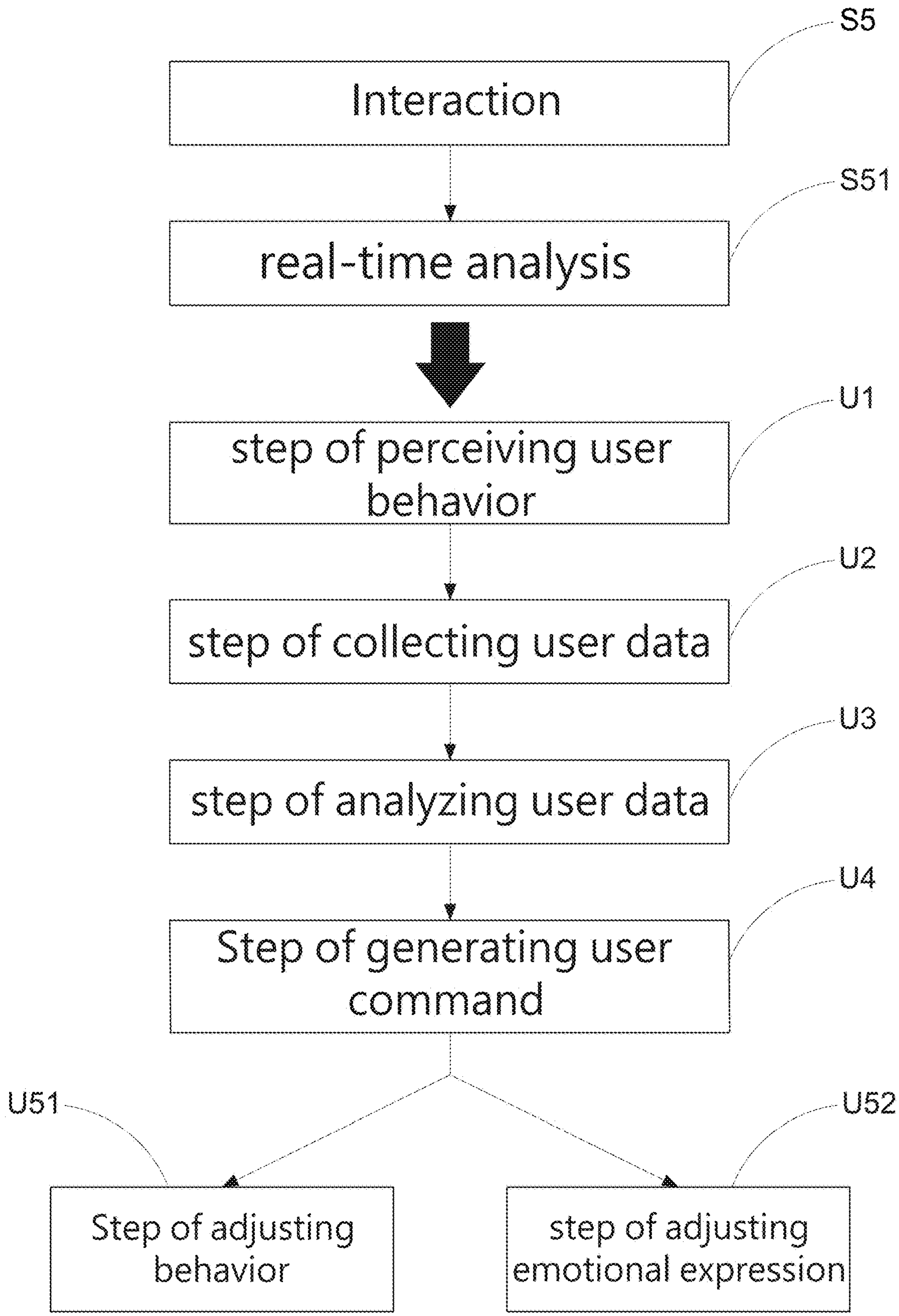

S5
Interaction
S51
real-time analysis
U1
step of perceiving user behavior
U2
step of collecting user data
U3
step of analyzing user data
U4
Step of generating user command
U51
Step of adjusting behavior
U52
step of adjusting emotional expression

Fig.6

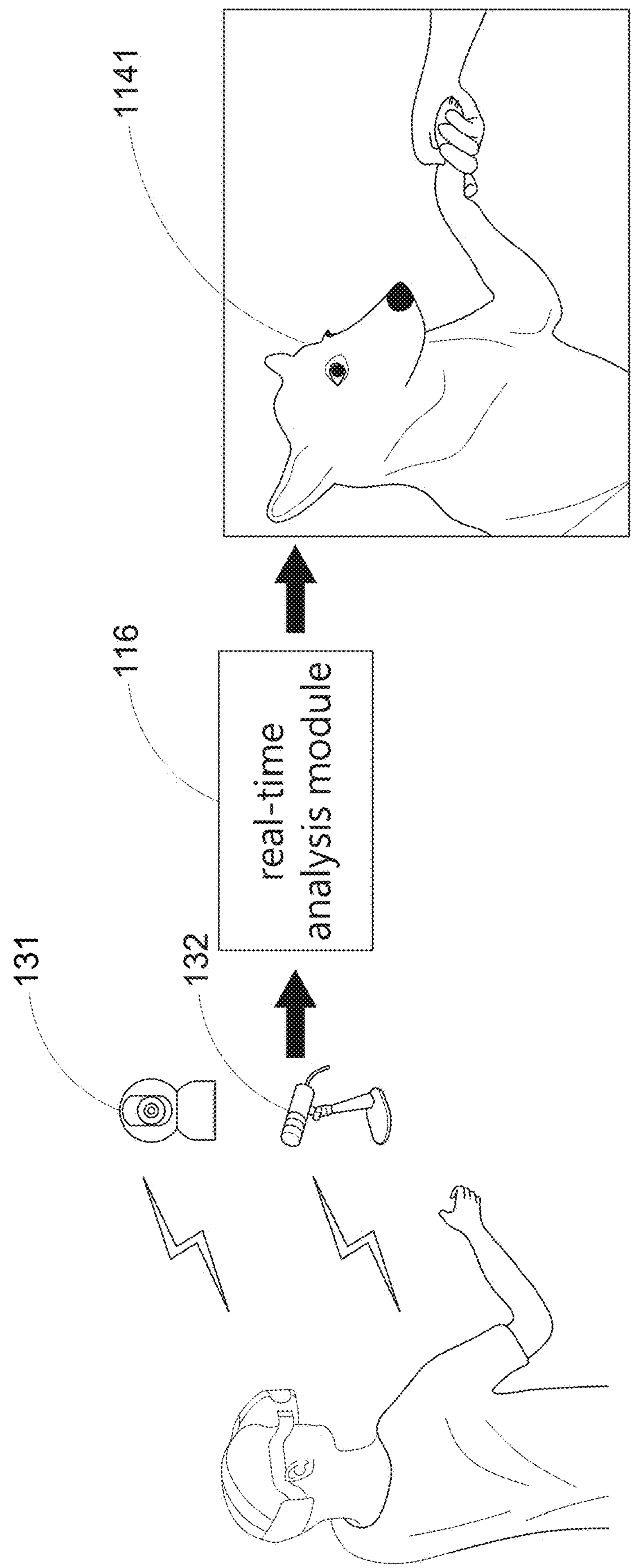


Fig.7

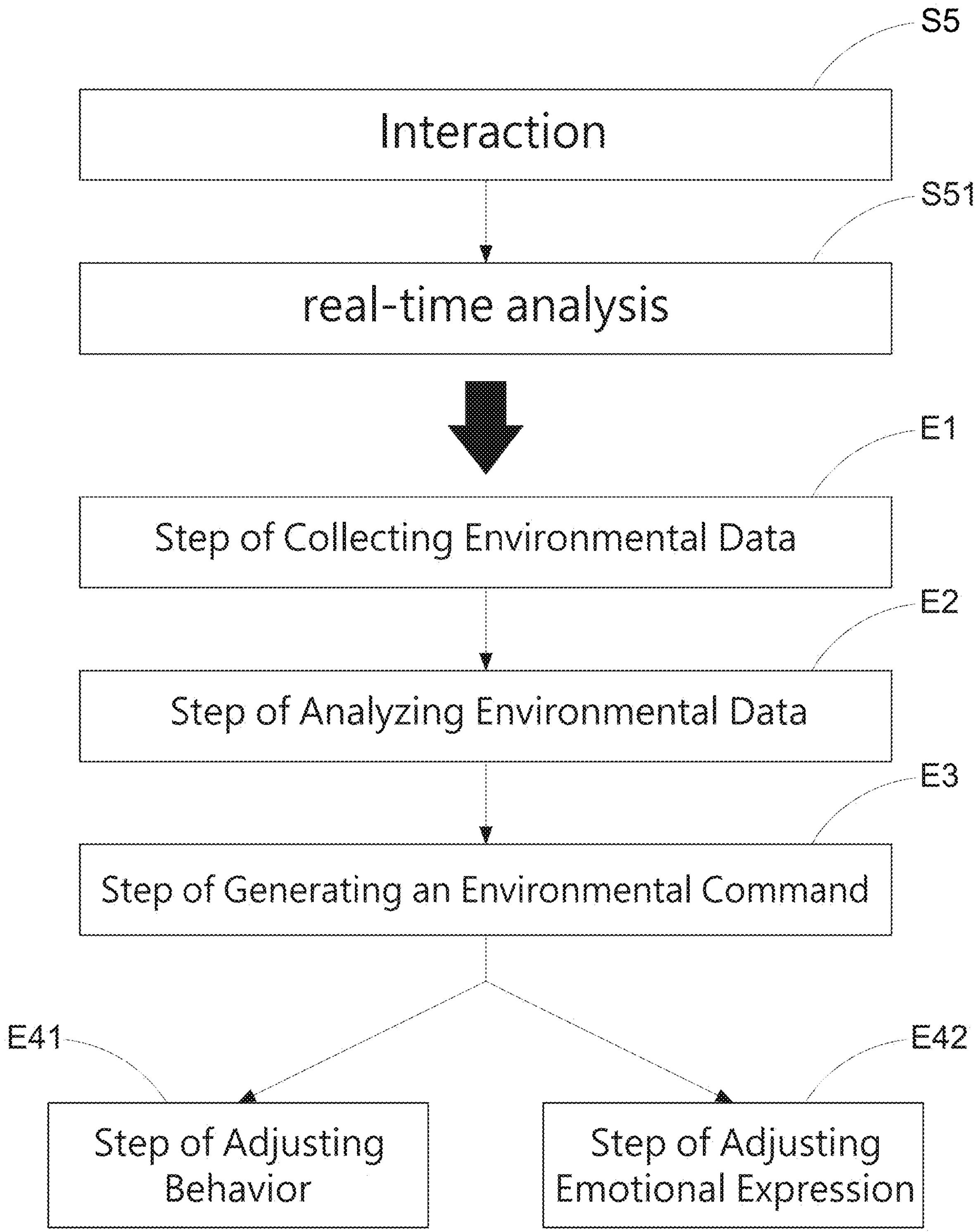

S5
Interaction
S51
real-time analysis
E1
Step of Collecting Environmental Data
E2
Step of Analyzing Environmental Data
E3
Step of Generating an Environmental Command
E41
Step of Adjusting Behavior
E42
Step of Adjusting Emotional Expression

Fig.8

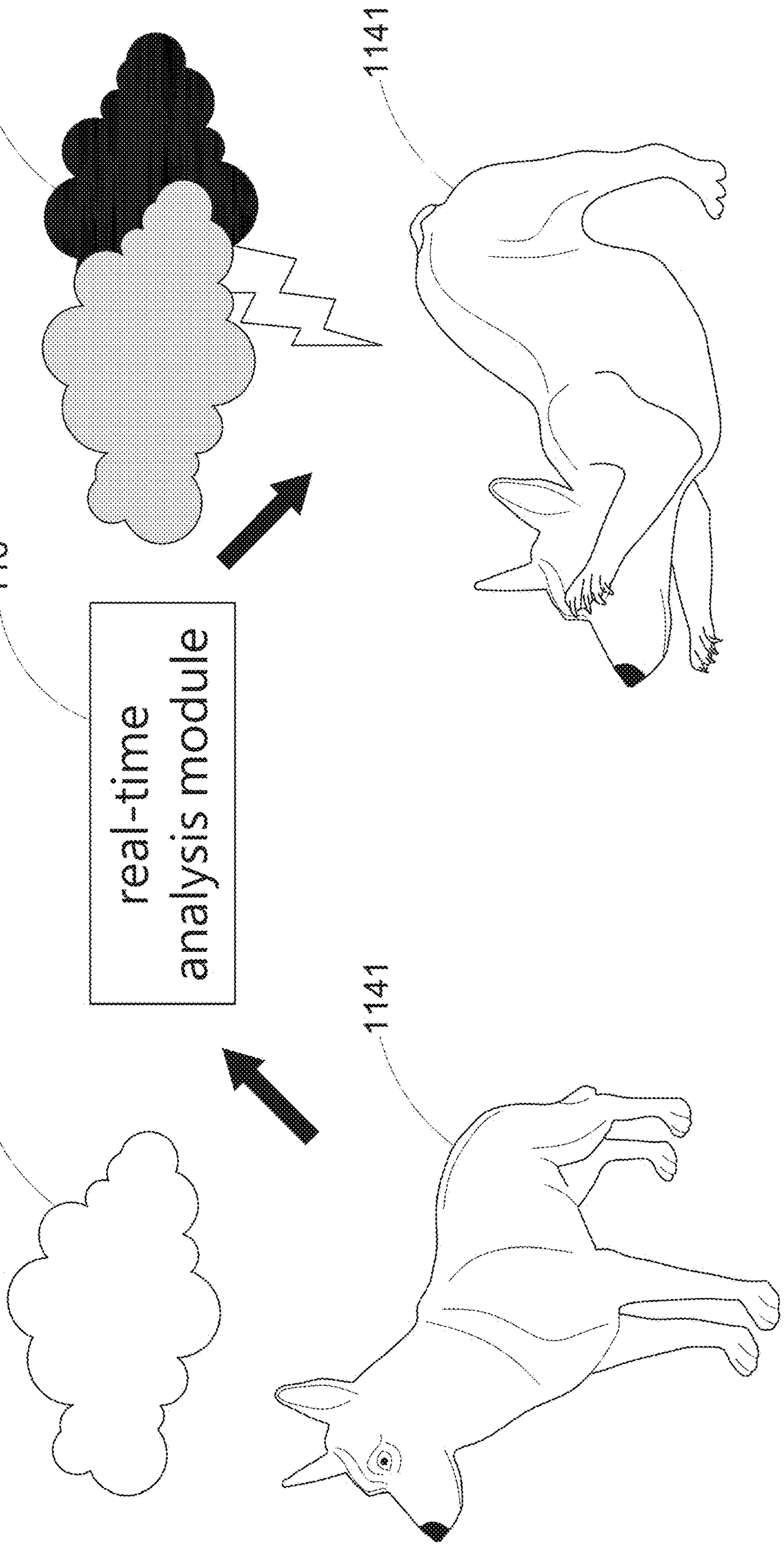


Fig.9

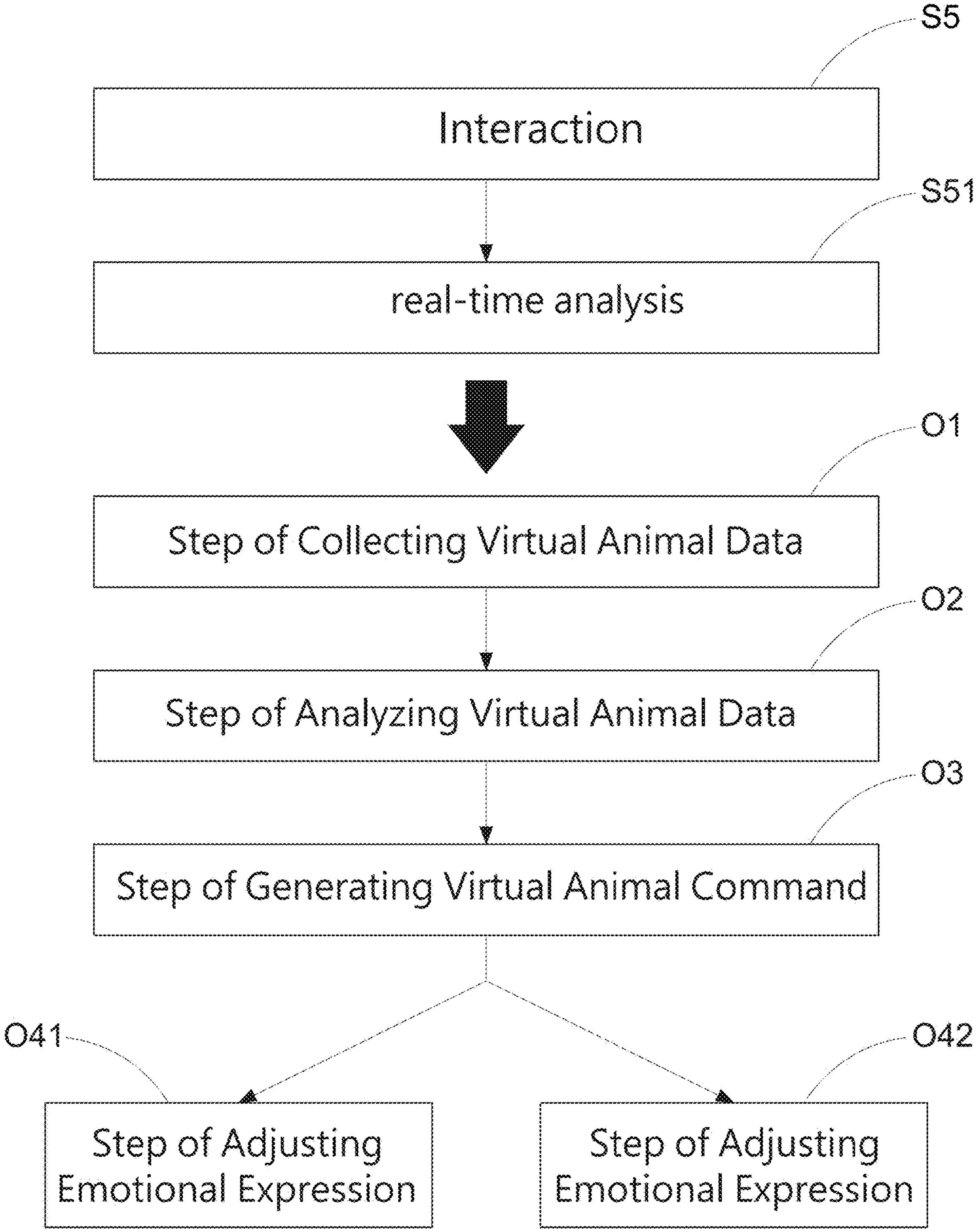

S5
Interaction
S51
real-time analysis
O1
Step of Collecting Virtual Animal Data
O2
Step of Analyzing Virtual Animal Data
O3
Step of Generating Virtual Animal Command
O41
Step of Adjusting Emotional Expression
O42
Step of Adjusting Emotional Expression

Fig.10

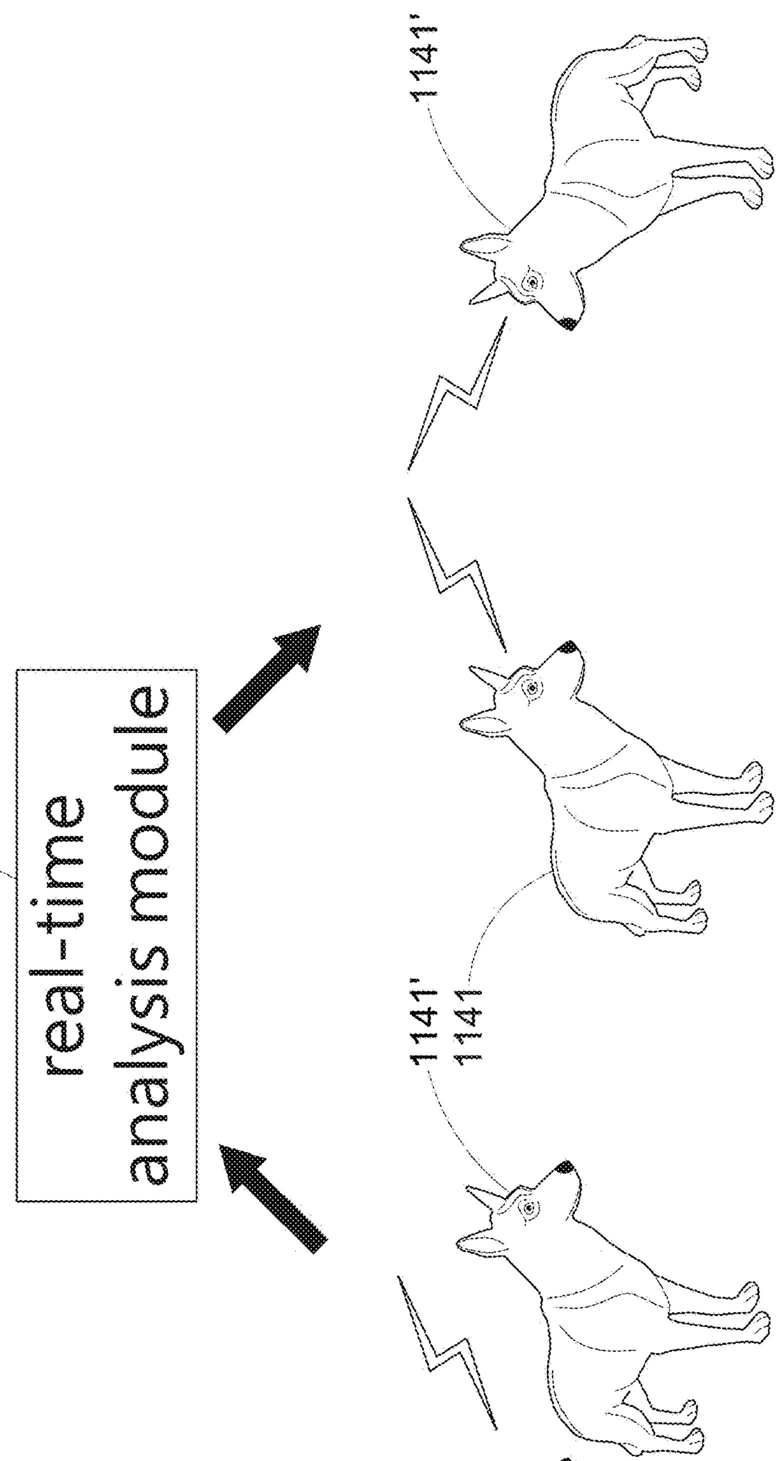


Fig.11

PET RECONSTRUCTION SYSTEM BASED ON A VIRTUAL WORLD INFRASTRUCTURE AND METHOD FOR IMPLEMENTING THEREOF

BACKGROUND OF INVENTION

(1) Field of the Present Disclosure

[0001] The present disclosure relates to a pet reconstruction system based on a virtual world infrastructure and a method for implementing thereof.

(2) Brief Description of Related Art

[0002] In the past, when a pet passed away, pet owners could only recall the time spent with their pets through videos and photos, which failed to offer an interactive experience. Traditional ways of remembering pets were limited to static visual and auditory elements, lacking dynamic, emotionally resonant interactions. Under such circumstances, pet owners may struggle to relive authentic moments with their pets. Moreover, this lack of interactivity in the remembrance process can create a sense of distance and helplessness for pet owners when reminiscing about their pets.

[0003] Chinese Patent Publication No. CN114870404A discloses a virtual pet generation method, apparatus, electronic device, and storage medium. The method includes obtaining multiple pet images captured from various angles by a user. Each pet image corresponds to a specific shooting angle. The method extracts features from the images to obtain characteristics of the pet's body parts. Based on these features, the method determines the virtual pet's appearance. Additionally, the method determines voice parameters for the virtual pet based on the user's characteristics. These voice parameters are used during interactions between the virtual pet and the user. Finally, the virtual pet is generated according to the appearance image and the voice parameters.

[0004] However, conventional virtual pet systems still lack precise control over specific behaviors and actions in different scenarios, such as the unique personality traits, habits, and movements of individual pets. In these systems, the behaviors of virtual pets are significantly limited, making it difficult to realistically simulate the rich variety of actions and reactions that pets might display in various environments. Therefore, a need exists to develop a virtual pet system that enables realistic interactions in diverse scenarios. Such a system would allow virtual pets to engage more authentically with users, the environment, and other virtual objects. It would provide pet owners with a way to recreate interactions with their pets in a virtual environment, enabling them to directly and dynamically interact with their deceased pets and cherish their memories in the virtual world.

SUMMARY OF INVENTION

[0005] It is a primary object of the present disclosure to provide to provide a pet reconstruction system and a method for reconstructing a pet in a virtual world infrastructure. This system allows a lifelike virtual pet to reappear before its owner and engage in interactive experiences, fulfilling the user's need for realistic interactions with a customized virtual pet and recreating the emotional connection between the owner and the pet in a virtual environment.

[0006] According to the present disclosure, a virtual world infrastructure comprises a server and one or more terminal devices. The server is in data communication with a model database, a customization database, a data processing module, a virtual pet generation module, an environment generation module, and a real-time analysis module. The model database stores multiple 3D appearance models, behavior models, and voice models for various animal species. The customization database stores the specific appearance features, specific behavior features, and specific voice features of a particular animal. The data processing module analyzes the specific features of the particular animal and compares them with the multiple models stored in the model database to generate a specific 3D appearance model, a specific behavior model, and a specific voice model for the particular animal, which are stored in the customization database. The virtual pet generation module creates a virtual identity for the particular animal based on its specific 3D appearance model, specific behavior model, and specific voice model to complete the reconstruction. The environment generation module creates a virtual environment and sends at least one environmental command to the real-time analysis module. The real-time analysis module receives at least one user command from one or more terminal devices and/or at least one environmental command from the environment generation module. Based on these commands, the virtual identity executes any one or a combination of its specific 3D appearance model, specific behavior model, and specific voice model. The terminal devices are in data communication with the server to download the virtual identity, including the specific 3D appearance model, specific behavior model, and specific voice model of the particular animal. This allows the virtual pet's behavior to be adjusted in real time, enabling it to exhibit more realistic responses in various environments. Consequently, the system satisfies the user's need for realistic interaction with a customized virtual pet and recreates the emotional connection between the owner and the pet in the virtual environment.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a block diagram illustrating the system architecture of the present disclosure;

[0008] FIG. **2** is the system architecture of the present disclosure;

[0009] FIG. **3** is a flow chart I of the present disclosure;

[0010] FIG. **4** is a flow chart II of the present disclosure;

[0011] FIG. **5** is a scenario diagram I of the present disclosure;

[0012] FIG. **6** is a flow chart III of the present disclosure;

[0013] FIG. **7** is a scenario diagram II of the present disclosure;

[0014] FIG. **8** is a flow chart IV of the present disclosure;

[0015] FIG. **9** is a scenario diagram III of the present disclosure;

[0016] FIG. **10** is a flow chart V of the present disclosure; and

[0017] FIG. **11** is a scenario diagram IV of the present disclosure.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0018] Referring to FIG. **1**, the pet reconstruction system **1** of the present disclosure is based on a virtual world infrastructure and is composed of a server **11** and one or more terminal devices **13**. The server **11** is in data communication with a model database **111**, a customization database **112**, a data processing module **113**, a virtual pet generation module **114**, an environment generation module **115**, and a real-time analysis module **116**. The model database **111** is used to store multiple 3D appearance models **1111**, multiple behavior models **1112**, and multiple voice models **1113** for various animal species. The customization database **112** is used to store specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** of a particular animal. The data processing module **113** analyzes the specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** of the specific animal and compares them with the multiple 3D appearance models **1111**, behavior models **1112**, and voice models **1113** stored in the model database **111**. Based on the comparison results, the data processing module **113** generates a specific 3D appearance model **1121'**, a specific behavior model **1122'**, and a specific voice model **1123'** for the specific animal, which are then stored in the customization database **112**. The virtual pet generation module **114** generates a virtual identity **1141** for the specific animal based on the specific 3D appearance model **1121'**, the specific behavior model **1122'**, and the specific voice model **1123'**, thereby completing the reconstruction. The environment generation module **115** generates a virtual environment **1151** and sends at least one environmental command to the real-time analysis module **116**. The real-time analysis module **116** receives at least one user command from one or more terminal devices **13** and/or at least one environmental command from the environment generation module **115**, causing the virtual identity **1141** to execute any one or a combination of the specific 3D appearance model **1121'**, the specific behavior model **1122'**, and the specific voice model **1123'** according to the received command. Additionally, the real-time analysis module **116** may further enable the virtual identity **1141** to send and/or receive at least one virtual pet command from at least one other virtual identity **1141'** in the virtual environment **1151**, causing the virtual identity **1141** to execute any one or a combination of the specific 3D appearance model **1121'**, the specific behavior model **1122'**, and the specific voice model **1123'** according to the virtual pet command. Moreover, one or more terminal devices **13** are in data-communication with the server **11** for downloading the virtual identity **1141**, which includes the specific 3D appearance model **1121'**, the specific behavior model **1122'**, and the specific voice model **1123'** of the specific animal.

[0019] The 3D appearance models **1111** stored in the model database **111** and the specific 3D appearance models **1121'** stored in the customization database **112** may respectively include, but are not limited to, at least one or a combination of the following parameters: breed, gender, age, skin color, fur color, markings, body size, and physical structure. The behavior models **1112** and the specific behavior models **1122'** may include at least one or a combination of the following parameters: sleep, foraging, communication, entertainment, learning, attack, defense, or other habits or actions. The voice models **1113** and the specific voice models **1123'** may respectively include, but are not limited to, at least one or a combination of the following parameters: pitch, frequency, volume, rhythm, continuity, and voice type.

[0020] The terminal device **13** may include a video camera **131** and a microphone **132**. The video camera **131** is used to capture real-time image data from the real world and transmit it to the server **11**. The microphone **132** is used to capture real-time audio data from the real world and transmit it to the server **11**. Additionally, the video camera **131** may be equipped with a motion sensing module **1311** for capturing motion data of at least one object within the real-time image data and transmitting it to the server **11**.

[0021] In the present disclosure, the real-time analysis module **116** may further adjust various parameters in the specific 3D appearance model **1121'**, the specific behavior model **1122'**, and the specific voice model **1123'** of the virtual identity **1141** based on environmental commands from the virtual environment **1151**. Additionally, the real-time analysis module **116** can adjust various parameters in the specific 3D appearance model **1121'**, the specific behavior model **1122'**, and the specific voice model **1123'** of the virtual identity **1141** based on real-time image data transmitted from the video camera **131** to the server **11**, real-time audio data transmitted from the microphone **132** to the server **11**, and/or motion data transmitted from the motion sensing module **1311** to the server **11**.

[0022] Referring to FIG. **2**, the server **11** can be in data communication with one or more terminal devices **13** via an Internet network **12**. Through the Internet **12**, users of the terminal devices **13** can download necessary data from the server **11** or upload required data to the server **11**. The Internet **12** may be a wide-area network, a 4G network, a 5G network, a Wi-Fi network, or a combination thereof. The terminal device **13** may be a mobile phone, a tablet computer, or a personal computer, among other devices capable of connecting to the server **11** via the network. Additionally, the terminal device **13** can be used in conjunction with a virtual reality display device **14**, which may be a glasses-type or head-mounted virtual reality display device for presenting any virtual object information downloaded to the terminal device **13**.

[0023] Referring to FIG. **3**, the method of implementation of the present disclosure includes the following steps:

[0024] 1. Data Upload Step S**1**: The specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** of a specific animal are uploaded to the customization database **112** of the server **11**.

[0025] 2. Model Construction Step S**2**: The data processing module **113** analyzes the specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** of the specific animal to generate the specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'**, which are then stored in the customization database **112**.

[0026] 3. Virtual Pet Generation Step S**3**: The virtual pet generation module **114** generates a virtual identity **1141** for the specific animal based on its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'**, thereby completing the reconstruction.

[0027] 4. Environment Generation Step S**4**: The environment generation module **115** generates a virtual environment **1151**.

[0028] 5. Interaction Step S**5**: The real-time analysis module **116** receives at least one user command from the terminal device **13** and/or at least one environmental command from the environment generation module **115**, causing the virtual identity **1141** to execute any one or a combination of the specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'**. Additionally, the real-time analysis module **116** may enable the virtual identity **1141** to send and/or receive at least one virtual pet command from at least one other virtual identity **1141'** in the virtual environment **1151**. Based on the virtual pet command, the virtual identity **1141** adjusts the parameters in the specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to perform corresponding actions.

[0029] The "upload" described in the Data Upload S**1** step may be implemented through various means. For example, a software application can be provided, allowing users to execute the application on the terminal device **13** and input the specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** of the specific animal. These features can then be uploaded to the customization database **112** of the server **11** via the Internet **12**. Alternatively, a web interface can be provided, enabling users to input the aforementioned features through a web browser and upload them to the customization database **112** of the server **11** via the Internet **12**. Furthermore, both the software application and the web interface may be equipped with an application programming interface API, allowing developers to write API requests using programming languages such as Python or Java. These API requests can upload the specified features in a particular format such as JSON or XML to the customization database **112** of the server **11**. In another example, the data containing the specified features may be stored on a conventional external storage device, such as a solid-state drive SSD or a hard disk drive HDD. The external storage device can then be directly connected to the physical equipment where the server **11** is located, enabling direct upload of the specified features to the customization database **112**.

[0030] Additionally, in the Data Upload S**1** step, an animal wearable device may be further provided, which may include but is not limited to an image sensor, such as a video camera, for capturing the specific appearance features **1121** of the specific animal; a behavior sensor, such as an accelerometer or gyroscope, for capturing the specific behavior features **1122** of the specific animal; and an audio sensor, such as a microphone, for capturing the specific voice features **1123** of the specific animal. The collected data can then be uploaded to the customization database **112** of the server **11** via the Internet **12**.

[0031] Referring to FIG. **4**, the Model Construction S**2** step may further include a "Model Optimization S**21**" step. In this step, the data processing module **113** first compares the specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** of the specific animal with multiple 3D appearance models **1111**, behavior models **1112**, and voice models **1113** of different animal species stored in the model database **111**. After the comparison, the data processing module **113** generates the specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** for the specific animal, thereby further improving the completeness and accuracy of these models.

[0032] Additionally, the Model Optimization S**21** step may utilize deep learning to achieve model optimization. For example, a generative adversarial network GAN can be employed, which includes the following steps:

[0033] 1. Data Preprocessing G**1** Step: This step involves preprocessing the specific appearance features **1121**, specific behavior features **1122**, and specific voice features **1123** collected in the Data Upload S**1** step, as well as other supplementary data e.g., text descriptions of the pet's behavior provided by the owner. Additionally, the multiple 3D appearance models **1111**, behavior models **1112**, and voice models **1113** stored in the model database **111** undergo data cleaning, normalization, and preprocessing to ensure data consistency. The processed data serves as real animal data.

[0034] 2. Generator Creation G**2** Step: A generator model is created to accept a random noise vector as input and output a predicted result for the species of the specific animal. For example, when the generator produces canine features based on a random noise vector, slightly unnatural predictions may appear in the generated output, such as irregularities in fur texture, color variations, or the shape of certain body parts.

[0035] 3. Discriminator Creation G**3** Step: A discriminator model is created to distinguish between real animal data and generated predictions. It identifies whether the input data is real animal data or generated predictions.

[0036] 4. GAN Model Training G**4** Step: The discriminator is trained using both real animal data and predictions generated by the generator. The discriminator learns to distinguish real animal data from generated predictions, while the generator is trained to deceive the discriminator such that the predictions become increasingly realistic.

[0037] 5. GAN Model Optimization G**5** Step: This step involves multiple iterations of training the GAN model in step G**4**. The generator and discriminator weights are adjusted as needed to optimize the performance of the GAN model.

[0038] Referring to FIG. **5**, when the provided specific features of the specific animal are incomplete, meaning that only partial feature data is uploaded, the comparison process in the Model Optimization S**21** step allows the data processing module **113** of the server **11** to search for available similar models in the model database **111** to supplement the missing specific features of the particular animal. Specifically, for example, if the provided feature data lacks the complete facial appearance, torso, limbs, voice, or inherent habits of the particular animal, the data processing module **113** can infer the missing features through the comparison process and use data from similar individuals of the same or related species in the model database **111** to further improve the completeness and accuracy of the model. This ensures that even when the specific feature data of the specific animal is insufficient, a more realistic virtual identity **1141** can still be established, providing users with a more comprehensive and lifelike experience.

[0039] Referring to FIG. **6**, in the Interaction S**5** step, a "Real-Time Analysis S**51**" step may be further included. In this step, the real-time analysis module **116** adjusts various parameters in the specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** of the virtual identity **1141** based on real-time image data transmitted from the video camera **131** to the server **11**, real-time audio data transmitted from the microphone **132** to the server **11**, and/or motion data transmitted from the motion sensing module **1311** to the server **11**. This adjustment enables the virtual identity **1141** to interact more closely with the user's behavior, providing the user with a more dynamic, personalized, and realistic experience.

[0040] Additionally, in the Real-Time Analysis S**51** step, the following steps may be further included following steps:

[0041] 1. Step U**1** of perceiving user behavior: The video camera **131**, microphone **132**, and/or motion sensing module **1311** detect the user's behavior, which may include a movement, a gesture, and/or a voice, either individually or in combination. The detected data is then transmitted to the server **11**.

[0042] 2. Step U**2** of collecting user data: The real-time analysis module **116** collects user-related data in the virtual environment **1151**, such as real-time user image data, real-time user audio data, and/or user motion data corresponding to the detected behavior.

[0043] 3. Step U**3** of analyzing user data: The real-time analysis module **116** performs emotional analysis on the collected user data to understand the user's intention and commands. For example, by using deep learning with a generative adversarial network GAN, the system can be trained with a large dataset containing emotional labels-such as positive, negative, or neutral emotions-based on facial expressions, voice intonation, or body gestures. This allows the real-time analysis module **116** to predict the emotional category associated with specific user actions or speech.

[0044] Step U**4** of generating user command: Based on the emotional analysis results, the real-time analysis module **116** generates a corresponding user command for the virtual identity **1141**.

[0045] Step U**51** of adjusting behavior: When the virtual identity **1141** receives the user command based on the emotional analysis results, it adjusts various parameters in its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to perform corresponding actions such as sleeping, foraging, or communicating. Alternatively, in the step U**52** of adjusting emotional expression, the virtual identity **1141** adjusts various parameters in its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to express corresponding emotions, such as excitement, fear, or happiness.

[0046] Referring to FIG. **7** in conjunction with FIG. **6**, when the user performs a handshake gesture, the real-time analysis module **116** can detect and process the user's visual action through the video camera **131** by capturing the motion and position of the user's hand to determine that the user is performing a handshake gesture. Alternatively, the real-time analysis module **116** may receive an audio signal through the microphone **132** to detect other possible verbal commands, such as a spoken handshake instruction. Subsequently, through the previously described steps U**1** to U**52**, the virtual identity **1141** performs the corresponding handshake action.

[0047] Referring to FIG. **8**, in the Real-Time Analysis S**51** step, the virtual identity **1141** may further adjust various parameters in the specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** based on environmental commands from the virtual environment **1151**, which includes the following steps:

[0048] Step E**1** of Collecting Environmental Data: The real-time analysis module **116** collects environmental change data from the virtual environment **1151**, including transitions between day and night, weather variations, lighting changes, and object movements.

[0049] Step E**2** of Analyzing Environmental Data: The real-time analysis module **116** conducts an environmental analysis on the collected environmental change data to determine trigger conditions. For instance, by utilizing deep learning with a generative adversarial network GAN, the system can be trained on a large dataset of environmental parameters, such as the intensity of thunder or rain sounds, the brightness of sunlight or lightning, or the magnitude of ground vibrations. This enables the real-time analysis module **116** to identify the appropriate behavior or emotional response of the virtual identity **1141** under specific environmental conditions.

[0050] Step E**3** of Generating an Environmental Command: Based on the results of the environmental analysis, the real-time analysis module **116** generates a corresponding environmental command for the virtual identity **1141**.

[0051] Step E**41** of Adjusting Behavior: Upon receiving the environmental command derived from the environmental analysis results, the virtual identity **1141** adjusts various parameters in its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to execute corresponding actions, such as sleeping, foraging, or communicating. Alternatively, in the Step E**42** of Adjusting Emotional Expression, the virtual identity **1141** modifies various parameters in its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to express corresponding emotions, such as excitement, fear, or happiness.

[0052] Referring to FIG. **9** in conjunction with FIG. **8**, when the environment generation module **115** introduces weather effects in the virtual environment **1151**, such as simulated dark clouds, lightning, and thunder sounds, the real-time analysis module **116** can, through the previously described steps E**1** to E**42**, trigger a fear emotion in the virtual identity **1141**. The virtual identity **1141** then performs a corresponding action by covering its eyes with its front paws.

[0053] Referring to FIG. **10**, in the Real-Time Analysis S**51** step, various parameters in specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** of the the virtual identity **1141** may be further adjusted based on the behavior and emotions of at least one other virtual identity **1141'** in the virtual environment **1151**. This includes the following steps:

[0054] Step O**1** of Collecting Virtual Animal Data: The real-time analysis module **116** collects behavioral and emotional data from other virtual identities **1141'** in the virtual environment **1151**, including sound, motion, and position data.

[0055] Step O**2** of Analyzing Virtual Animal Data: The real-time analysis module **116** performs a virtual animal analysis on the collected behavioral and emotional data of the other virtual identities **1141'** to determine trigger conditions. For example, by utilizing deep learning with a generative adversarial network GAN, the system can be trained on a large dataset containing animal interaction data, such as mutual sniffing, playing, or coordinated actions. This enables the real-time analysis module **116** to identify the specific behavior or emotional response that should be triggered in the virtual identity **1141** based on the behavior and emotions of the other virtual identities **1141'**.

[0056] Step O**3** of Generating a Virtual Animal Command: Based on the results of the virtual animal analysis, the real-time analysis module **116** generates a corresponding virtual animal command for the virtual identity **1141**.

[0057] Step O**41** of Adjusting Behavior: Upon receiving the virtual animal command derived from the virtual animal analysis results, the virtual identity **1141** adjusts various parameters in its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to execute corresponding actions, such as playing, foraging together, or communicating. Alternatively, in the Step O**42** of Adjusting Emotional Expression, the virtual identity **1141** modifies various parameters in its specific 3D appearance model **1121'**, specific behavior model **1122'**, and specific voice model **1123'** to express corresponding emotions, such as excitement, fear, or happiness.

[0058] Referring to FIG. **11** in conjunction with FIG. **10**, when a specific signal is sent to the virtual identity **1141** by a neighboring virtual identity **1141'**, the real-time analysis module **116**, through the previously described steps O**1** to O**42**, may trigger a social behavior in the virtual identity **1141** and cause it to respond accordingly.

[0059] From the above, it is evident that the present invention, once implemented, can indeed achieve the objective of providing a pet reconstruction system and a method for reconstructing a pet in a virtual world. This system is based on a virtual world infrastructure that allows a lifelike virtual pet to reappear before its owner and engage in interactive experiences. It fulfills the user's need for realistic interactions with a customized virtual pet, thereby recreating the emotional connection between the owner and the pet in the virtual environment.

[0060] It should be noted that the above descriptions are merely preferred embodiments of the present invention and are not intended to limit the scope of its implementation. Any equivalent modifications and variations made by those skilled in the art without departing from the spirit and scope of the invention shall fall within the scope of the claims of the present invention.

LABELS IN GRAPHICS

[0061] **1** pet reconstruction system
[0062] **11** server
[0063] **111** model database
[0064] **112** customization database
[0065] **113** data processing module
[0066] **114** virtual pet generation module
[0067] **115** environment generation module
[0068] **116** real-time analysis module
[0069] **1111** 3D appearance model
[0070] **1112** multiple behavior model
[0071] **1113** multiple voice model
[0072] **1121** specific appearance feature
[0073] **1122** specific behavior feature
[0074] **1123** specific voice feature
[0075] **1121'** specific 3D appearance model
[0076] **1122'** specific behavior model
[0077] **1123'** specific voice model
[0078] **1141** virtual identity
[0079] **1141'** other virtual identity
[0080] **1151** virtual environment
[0081] **12** Internet
[0082] **13** terminal device
[0083] **131** video camera
[0084] **132** microphone
[0085] **1311** motion sensing module
[0086] **14** virtual reality display device

What is claimed is:

1. A pet reconstruction system based on a virtual world infrastructure, wherein the system is built on a virtual world infrastructure and primarily comprises a server and one or more terminal devices, the server comprising:

a model database, in data communication with the server, configured to store multiple 3D appearance models, multiple behavior models, and multiple voice models for various animal species;

a customization database, in data communication with the server, configured to store uploaded specific appearance features, specific behavior features, and specific voice features of a specific animal;

a data processing module, in data communication with the server, configured to analyze the specific appearance features, specific behavior features, and specific voice features of the specific animal to generate a specific 3D appearance model, a specific behavior model, and a specific voice model for the specific animal, and store the generated models in the customization database;

a virtual pet generation module, in data communication with the server, configured to generate a virtual identity for the particular animal based on the specific 3D appearance model, specific behavior model, and specific voice model of the specific animal to complete the reconstruction; and

a real-time analysis module, in data communication with the server, configured to receive at least one user command from the terminal device and cause the virtual identity to execute any one or a combination of the specific 3D appearance model, specific behavior model, and specific voice model according to the user command, thereby adjusting various parameters in the specific 3D appearance model, specific behavior model, and specific voice model to perform a corresponding action.

2. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **1**, wherein the terminal device is in data communication with the server via the Internet for downloading the virtual identity, which includes the specific 3D appearance model, specific behavior model, and specific voice model of the specific animal.

3. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **1**, wherein the terminal device further comprises:

a video camera for capturing real-time image data from the real world; and

a microphone for capturing real-time audio data from the real world, wherein the video camera includes a motion sensing module for capturing motion data of at least one object within the real-time image data and transmitting the motion data to the server.

4. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **3**, wherein the server further comprises an environment generation module for generating a virtual environment and transmitting at least one environmental command to the real-time analysis module.

5. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **4**, wherein the real-time analysis module receives the environmental command from the environment generation module and causes the virtual identity to execute any one or a combination of the specific 3D appearance model, specific behavior model, and specific voice model according to the environmental command.

6. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **4**, wherein the real-time analysis module further enables the virtual identity to send and/or receive at least one virtual animal command from at least one other virtual identity in the virtual environment and causes the virtual identity to execute any one or a combination of the specific 3D appearance model, specific behavior model, and specific voice model according to the virtual animal command.

7. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **4**,

wherein each 3D appearance model and specific 3D appearance model respectively include at least one or a combination of the following parameters: breed, gender, age, skin color, fur color, markings, body size, and physical structure;

wherein each behavior model and specific behavior model respectively include at least one or a combination of the following parameters: sleeping, foraging, communicating, entertaining, learning, attacking, and defending; and

wherein each voice model and specific voice model respectively include at least one or a combination of the following parameters: pitch, frequency, volume, rhythm, continuity, and voice type.

8. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **7**, wherein the real-time analysis module further adjusts the parameters in the specific 3D appearance model, specific behavior model, and specific voice model according to the environmental command.

9. The pet reconstruction system based on a virtual world infrastructure as claimed in claim **7**, wherein the real-time analysis module further adjusts the parameters in the specific 3D appearance model, specific behavior model, and specific voice model based on the real-time image data transmitted from the video camera, real-time audio data transmitted from the microphone, and/or motion data transmitted from the motion sensing module to the server.

10. A method for implementing a pet reconstruction system based on a virtual world infrastructure, comprising:

a data upload step, in which specific appearance features, specific behavior features, and specific voice features of a specific animal are uploaded to a customization database of a server;

a model construction step, in which a data processing module of the server analyzes the specific appearance features, specific behavior features, and specific voice features of the specific animal to generate a specific 3D appearance model, a specific behavior model, and a specific voice model for the specific animal, and stores the generated models in the customization database;

a virtual pet generation step, in which a virtual pet generation module of the server generates a virtual identity for the specific animal based on the specific 3D appearance model, specific behavior model, and specific voice model;

an environment generation step, in which an environment generation module of the server generates a virtual environment; and

an interaction step, in which a real-time analysis module of the server receives at least one user command from one or more terminal devices in data communication with the server and/or at least one environmental command from the environment generation module, causing the virtual identity to execute any one or a combination of the specific 3D appearance model, specific behavior model, and specific voice model according to the user command and/or the environmental command.

11. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **10**, wherein the terminal device further comprises a video camera for capturing real-time image data from the real world; a microphone for capturing real-time audio data from the real world; and the video camera may include a motion sensing module for capturing motion data of at least one object within the real-time image data and transmitting the motion data to the server.

12. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **11**, wherein the terminal device further comprises a video camera for capturing real-time image data from the real world; a microphone for capturing real-time audio data from the real world; and the video camera may include a motion sensing module for capturing motion data of at least one object within the real-time image data and transmitting the motion data to the server.

13. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **12**, wherein the model optimization step uses a generative adversarial network (GAN).

14. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **12**, wherein each 3D appearance model and specific 3D appearance model respectively include at least one or a combination of the following parameters: breed, gender, age, skin color, fur color, markings, body size, and physical structure; each behavior model and specific behavior model respectively include at least one or a combination of the following parameters: sleeping, foraging, communicating, entertaining, learning, attacking, and defending; and each voice model and specific voice model respectively include at least one or a combination of the following parameters: pitch, frequency, volume, rhythm, continuity, and voice type.

15. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **14**, wherein the interaction step further includes a real-time analysis step, in which the real-time analysis module adjusts the parameters in the specific 3D appearance model, specific behavior model, and specific voice model based on real-time image data transmitted from the video camera, real-time audio data transmitted from the microphone, and/or motion data transmitted from the motion sensing module to the server.

16. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **15**, wherein the real-time analysis step further comprises:

a step of perceiving user behavior, in which the video camera, the microphone, and/or the motion sensing module detect a movement, a gesture, and/or a voice and transmit the detected data to the server;

a step of collecting user data, in which the real-time analysis module collects real-time user image data, real-time user audio data, and/or user motion data corresponding to the detected movement, gesture, and/or voice;

a step of analyzing user data, in which the real-time analysis module performs an emotional analysis on the collected real-time user image data, real-time user audio data, and/or user motion data;

a step of analyzing user data, in which the real-time analysis module performs an emotional analysis on the collected real-time user image data, real-time user audio data, and/or user motion data;

a step of generating user command, in which the real-time analysis module generates a corresponding user command for the virtual identity based on the emotional analysis results; and

a step of adjusting behavior and/or adjusting emotional expression, in which the virtual identity adjusts various parameters in the specific 3D appearance model, specific behavior model, and specific voice model after receiving the user command based on the emotional analysis results.

17. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **15**, wherein the real-time analysis step further comprises:

a step of collecting environmental data, in which, in which the real-time analysis module collects environmental change data from the virtual environment;

a step of analyzing environmental data, in which the real-time analysis module performs an environmental analysis on the collected environmental change data;

a step of generating an environmental command, in which the real-time analysis module generates a corresponding environmental command for the virtual identity based on the results of the environmental analysis; and

a step of adjusting behavior and/or expressing emotions, in which the virtual identity adjusts various parameters in the specific 3D appearance model, specific behavior model, and specific voice model after receiving the environmental command based on the results of the environmental analysis.

18. The method for implementing a pet reconstruction system based on a virtual world infrastructure as claimed in claim **15**, wherein the real-time analysis step further comprises:

a step of collecting virtual animal data, in which the real-time analysis module collects behavioral and emotional data of at least one other virtual identity within the virtual environment;

a step of analyzing virtual animal data, in which the real-time analysis module performs a virtual animal analysis on the collected behavioral and emotional data of the other virtual identity;

a step of generating a virtual animal command, in which the real-time analysis module generates a corresponding virtual animal command for the virtual identity based on the results of the virtual animal analysis; and

a step of adjusting behavior and/or a step of adjusting emotional expression, in which the virtual identity modifies various parameters in the specific 3D appearance model, specific behavior model, and specific voice model after receiving the virtual animal command based on the results of the virtual animal analysis.

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